Semester Plan

Introduction to Machine Learning Fall 2026

#	Date	Title	Topics
1	2026-09-03	Foundations & Parametric Classifiers I	Jupyter/Python setup; classification vs. regression; single-layer perceptron; train/validation/test split; confusion matrix, precision, recall, ROC curves
2	2026-09-01	Parametric Classifiers II — Neural Networks	Multilayer perceptrons; activation functions (sigmoid, ReLU, tanh); backpropagation; gradient descent; performance comparison vs. single-layer perceptron
3	2026-09-17	Parametric Regression I	Linear regression; R^2; fit percentage; worked example (linear model fit to nonlinear data); intro to polynomial regression
4	2026-09-24	Parametric Regression II	Logistic regression; polynomial regression; regularisation (ridge, lasso); model comparison across regression types; non-linear regression (kernel regression, regression trees)
5	2026-10-08	Non-Parametric Classifiers	Feature standardisation; K-nearest neighbour; classification trees; random forests (ensemble method, bagging); parametric vs. non-parametric comparison
6	2026-10-22	Kernel Methods I	Kernel trick; kernel-based perceptron; intro to Support Vector Machines
7	2026-10-29	Gaussian Processes	Gaussian Processes: prior/posterior over functions; GP regression; uncertainty quantification; robotics relevance (sensor models, safe exploration)
8	2026-11-05	Kernel Methods II — SVM & Loss Functions	SVM dual problem; (optional) interior point methods; multi-class SVM (one-vs-rest, one-vs-one); loss functions; optimisation methods recap
9	2026-11-12	Supervised vs. Unsupervised Learning	Supervised vs. unsupervised framing; K-means clustering (init sensitivity); elbow method (number of clusters); Gaussian Mixture Models
10	2026-11-19	Generative vs. Discriminative Models	Discriminative review (logistic regression, SVM, perceptron); generative model concept; Naive Bayes
11	2026-11-26	Classical Planning & Search	Search; A* search; behaviour trees; knowledge representation
12	2026-12-03	Course Recap & Q&A	Course synthesis: discriminative vs. generative vs. decision-making recap; key trade-offs (parametric vs. non-parametric, model-based vs. learned); open Q&A